TECHNISCHE UNIVERSITÄT
ILMENAU

Fakultät für Wirtschaftswissenschaften und Medien

Dr.-Ing..
Schwandt, Tobias

Fachgebiet Virtuelle Welten und Digitale Spiele
Fakultät für Wirtschaftswissenschaften und Medien

Technische Universität Ilmenau
Ernst Abbe Zentrum für Forschung und Transfer (EAZ)
Ehrenbergstr. 29
98693 Ilmenau

tobias.schwandt@tu-ilmenau.de
www.tu-ilmenau.de/vwds

Muhammad Abid Zahid

Technische Universität Ilmenau | PF 10 05 65 | 98684 Ilmenau

Second report on the master thesis:

AI-Guided Voice-Interactive VRMeditation: EnhancingRelaxation through Adaptive Audiovisual Cues and Biofeedback

submitted by Muhammad Abid Zahid (63677) on 06. March 2026

Notes about the Expertise:

The ratings shown in the table below do not correspond to the grading scale. In particular, the categories are not equally important. Particularly good but also particularly bad performance on individual points can strongly influence the overall impression of a work. It is generally not possible to calculate an average value with the evaluations and derive the final grade from this. The individual dimensions only serve to make it clear in which good and in which less good performance was achieved.

Overview

Dimension	Criteria	- -	-	+/-	+	++
(1) Topic / RQ	The question and goal of the work were formulated pointedly and precisely.				X	
	The relevance of the topic was clearly identified.			X		
	The claim and scope of the formulated objective is reasonable (no deviation from the agreed objective).				X	
	Comment: Mr. Zahid has chosen a compelling and highly relevant subject for his master's thesis: the role of meditation in fostering physical and, more importantly, mental well-being. In a world where stress is ubiquitous, meditation offers a promising pathway to relief. His research question asks whether virtual reality can enhance the effectiveness of meditation practices. This inquiry is highly motivated by its potential to advance both our theoretical understanding and practical applications in mental health care.					
(2) Fundamentals & state-of-the-art	The necessary fundamentals for understanding the work are described.					X
	The state-of-the-art on the issue was presented in an up-to-date and internationally structured manner.				X	
	The relevant approaches and procedures are presented to an appropriate extent.				X	
	The chosen method(s) is/are justified from the presentation of the state-of-the-art.				X	
	Comment: He presents a comprehensive review of contemporary research, employing state-of-the-art methodologies that establish a solid foundation for his own approaches—including the development of novel methods, systematic comparisons, and robust conclusions. However, a handful of related studies remain unexamined and are not benchmarked against his results, limiting the full contextualization of his contributions.					
(3) Methodology / Implementation	The methodology carried out is discussed and justified appropriately.			X		
	The methodology is based on current state-of-the-art procedures.				X	
	The methodology reveals new findings or confirms or refutes existing assumptions.				X	
	The own contribution inside the work is clearly emphasized.			X		
	The tools used to understand the methodology are adequately described.					X
	Comment: The research design is articulated clearly and transparently. However, the methodology section is dominated by a highly technical description that merges implementation details with experimental procedures. In practice, the specific coding or hardware steps are less critical than the features and capabilities of the prototype itself. To improve clarity, the manuscript should shift focus toward a more detailed account of how the AI components and audiovisual assets are integrated into the system. Moreover, the current description omits a clear outline of the meditation protocols employed for both the conventional and VR user groups. Providing a step-by-step account of the meditation sessions—duration, guidance style, sensor feedback, and any pre- or post-session measures—would greatly strengthen the methodological rigor and reproducibility of the study.					
(4) Results, Evaluation & Discussion	The presentation of results is appropriately structured.			X		
	Results are presented in appropriate graphics/tables with text descriptions.		X			
	Appropriate interpretation of the results: No overinterpretation no underinterpretation.			X		
	Significant overall discussion & conclusion/innovative outlook.			X		
	Comment: The results are presented clearly and are underpinned by appropriate significance tests, which lend statistical robustness to the findings. Nonetheless, the discussion section is somewhat terse. Extending the interpretation to explore the broader implications of the results, considering theoretical and practical ramifications, would					

	substantially strengthen this part of the manuscript. While the research questions are addressed at the conclusion, the evidence gathered seems insufficient to fully answer RQ2 and RQ3. Acknowledging this limitation—and proposing how future studies might fill the gap—would provide a more balanced and honest appraisal of the work.					
(5) Formalities	Clear overall structure and chapter structure of the work.					X
	Correct citations and correct bibliography.					X
	Correct abstract.				X	
	Easy to read scientific language style.					X
	Correct spelling, grammar.				X	
	Correct layout.					X
	Comment: The master thesis is well-written.					

Possible questions for the defense:

1. What was the effect of VR and AI-generated voice cues on users' meditation experiences?
2. Why was a solid-color background chosen for the visualization instead of a conventional skybox?
3. Which specific within-subject design elements were employed in your user study, and could you provide a more detailed explanation?
4. What was the rationale for converting the biofeedback data to a LibreOffice-compatible format?
5. Why does page 31 of the thesis show missing questionnaire responses?
6. Why were participants who practiced conventional meditation not asked to report their stress levels?
7. How did you measure the impact of the closed-eye visuals, and what data were obtained to support this measurement?

Ilmenau, 06.03.2026

Tobias Schwandt